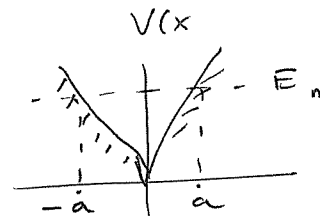


Solution set 10

Problem 1 we want to find the bound state energies. Let the energy be E_n . Then we know from the WKB formula.



$$\int_{-a}^a P(x) dx = \left(n + \frac{1}{2}\right) \pi \hbar \quad n=0, 1, 2, \dots$$

$$V(x) = k|x|$$

note: $a = \frac{E_n}{k}$

where
$$P(x) = \sqrt{2m(E_n - V(x))}$$
$$= \sqrt{2m(E_n - k|x|)}$$

$$\Rightarrow 2 \int_0^a dx \sqrt{2m(E_n - kx)} = \left(n + \frac{1}{2}\right) \pi \hbar$$

$$\Rightarrow 2 \int_0^a dx \sqrt{2mk(a-x)} = \left(n + \frac{1}{2}\right) \pi \hbar$$

$$\Rightarrow 2\sqrt{2mk} \int_0^a dx \sqrt{a-x} = \left(n + \frac{1}{2}\right) \pi \hbar$$

$$\Rightarrow 2\sqrt{2mk} \int_0^a dy \sqrt{y} = \left(n + \frac{1}{2}\right) \pi \hbar$$

Note:
$$\int_0^a dx \sqrt{a-x} = \int_0^a dy \sqrt{y} = \frac{2}{3} a^{3/2}$$

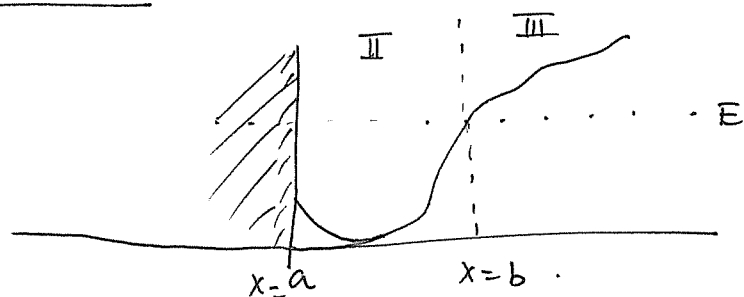
$$\Rightarrow 2\sqrt{2mk} \frac{2}{3} a^{3/2} = \left(n + \frac{1}{2}\right) \pi \hbar$$

$$\Rightarrow a = \left\{ \frac{3}{4\sqrt{2}} \left(n + \frac{1}{2}\right) \pi \frac{\hbar}{\sqrt{mk}} \right\}^{2/3}$$

$$\Rightarrow E_n = \left\{ \frac{3}{4\sqrt{2}} \left(n + \frac{1}{2}\right) \pi \cdot \frac{\hbar k}{\sqrt{m}} \right\}^{2/3}$$

$$n = 0, 1, 2, \dots$$

Problem 2 : The potential $V(x)$ is of the form



We know $\psi(x) = 0$ for $x \leq a$ Since $V(x) = \infty$ there.

Thus in the notation of the class we only have two regions region II and region III. (Note a, b are reversed now w.r.t the class!)

In the WKB approx. the solutions in these two regions is of the form

Region III

$$\psi(x) = \frac{A}{\sqrt{q(x)}} \exp\left(-\frac{i}{\hbar} \int_b^x q(x) dx\right) + \frac{B}{\sqrt{q(x)}} \exp\left(\frac{i}{\hbar} \int_b^x q(x) dx\right)$$

Region II

$$\psi(x) = \frac{C}{\sqrt{p(x)}} \exp\left(\frac{i}{\hbar} \int_x^b p(x) dx\right) + \frac{D}{\sqrt{p(x)}} \exp\left(-\frac{i}{\hbar} \int_x^b p(x) dx\right)$$

With the connection formula

$$C = \left(A - \frac{B}{2i}\right) e^{-i\pi/4}$$

$$D = \left(A + \frac{B}{2i}\right) e^{i\pi/4}$$

For the bound state we need $B = 0$. Further we also need that $\psi(x=a) = 0$. That is the WKB solution in region II should vanish at $x=a$.

we get

$$\frac{C}{\sqrt{p(a)}} \exp\left(\frac{i}{\hbar} \int_a^b p(x) dx\right) + \frac{D}{\sqrt{p(a)}} \exp\left(-\frac{i}{\hbar} \int_a^b p(x) dx\right) = 0$$

From the connection formula this means

$$\frac{A}{\sqrt{p(a)}} \left\{ \exp\left\{\frac{i}{\hbar} \int_a^b p(x) dx - i\frac{\pi}{4}\right\} + \exp\left\{-\frac{i}{\hbar} \int_a^b p(x) dx + i\frac{\pi}{4}\right\} \right\} = 0$$

$$\Rightarrow \frac{2A}{\sqrt{p(a)}} \cos\left[\frac{1}{\hbar} \int_a^b p(x) dx - \frac{\pi}{4}\right] = 0$$

which means

$$\frac{1}{\hbar} \int_a^b p(x) dx - \frac{\pi}{4} = (2n+1)\frac{\pi}{2} \quad n = 0, 1, 2, \dots$$

$$\boxed{\int_a^b p(x) dx = \left(n + \frac{3}{4}\right) \pi \hbar}$$

Note: The condition now is different from the earlier condition for usual potentials!

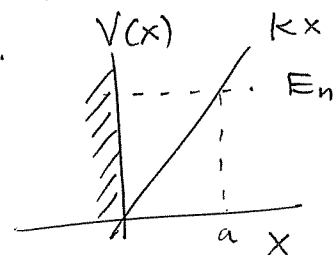
Now if the potential is given by

we get

$$\int_0^a dx \sqrt{2mk(a-x)} = \left(n + \frac{3}{4}\right) \pi \hbar$$

which then leads to (following steps from problem 1).

$$E_n = \left\{ \frac{3}{2\sqrt{2}} \left(n + \frac{3}{4}\right) \pi \frac{\hbar k}{\sqrt{m}} \right\}^{2/3} \quad n = 0, 1, 2, \dots$$



Important: Note you could have guessed this from the solution to problem 1. Since the potential in problem 1 is even the eigenfunctions are "even" and "odd".
 (n=0,2,...) (n=1,3,...). When the potential is $V(x) = \infty$ $x < 0$ then only the odd solutions survive so

$$E_n = \left\{ \frac{3}{4\sqrt{2}} \left(n + \frac{1}{2} \right) \pi \cdot \frac{\hbar k}{\sqrt{m}} \right\}^{2/3} \quad n = 1, 3, 5, \dots$$

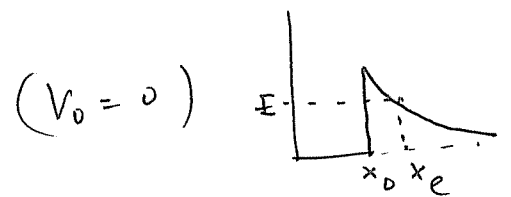
$$\Rightarrow E_{n'} = \left\{ \frac{3}{4\sqrt{2}} \left(2n' + \frac{3}{2} \right) \pi \cdot \frac{\hbar k}{\sqrt{m}} \right\}^{2/3} \quad n' = 0, 1, 2, \dots$$

$$E_{n'} = \left\{ \frac{3}{2\sqrt{2}} \left(n' + \frac{3}{4} \right) \pi \cdot \frac{\hbar k}{\sqrt{m}} \right\}^{2/3} \quad n' = 0, 1, 2, \dots$$

(as found earlier!).

Problem 3: The mean life time is given by $\tau = \frac{1}{R}$ where

$$R = \frac{\sqrt{2m(E+V_0)}}{2mx_0} e^{-\gamma}$$



$$\gamma = \frac{2}{\hbar} \int_{x_0}^{x_e} \sqrt{2m} (V(x) - E)^{1/2} dx$$

In the range x_0 to x_e $V(x) = \frac{2Ze^2}{x}$

(α -particles have charge $2e$)
 (Nucleus has charge Ze).

$$E = \frac{2Ze^2}{x_e}$$

Thus

$$\gamma = \frac{2}{\hbar} \int_{x_0}^{x_e} \sqrt{2m} \sqrt{2Ze^2} \left(\frac{1}{x} - \frac{1}{x_e} \right)^{1/2} dx$$

$$r = \frac{2}{\hbar} \sqrt{4m(Ze^2)} \int_{x_0}^{x_e} dx \left(\frac{1}{x} - \frac{1}{x_e} \right)^{1/2}$$

Set $\frac{x}{x_e} = t$, $y = \frac{x_0}{x_e}$, $\Rightarrow dt = \frac{dx}{x_e}$,

$$r = \frac{2}{\hbar} \sqrt{4m(Ze^2)} x_e \int_y^1 \frac{dt}{\sqrt{t}} \sqrt{1-t}$$

Set $\sqrt{t} = \sin \theta$

$$\int_y^1 \frac{dt}{\sqrt{t}} \sqrt{1-t} = 2 \int_{\sin^{-1}\sqrt{y}}^{\pi/2} d\theta \cos^2 \theta = \int_{\sin^{-1}\sqrt{y}}^{\pi/2} d\theta (1 + \cos 2\theta)$$

$$= \left[\cos^{-1} \sqrt{y} - \sqrt{y(1-y)} \right]$$

Thus

$$r = \frac{2}{\hbar} \sqrt{4m x_e Z e^2} \left(\cos^{-1} \sqrt{y} - \sqrt{y(1-y)} \right)$$

The formula for r simplifies when written in terms of the velocity inside the well

$$v = \frac{\sqrt{2mE}}{m} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{4Ze^2}{m x_e}}$$

$$\Rightarrow \sqrt{x_e} = \sqrt{\frac{4Ze^2}{m}} \cdot \frac{1}{v}$$

Thus

$$r = \frac{8(Ze^2)}{\hbar v} \left(\cos^{-1} \sqrt{y} - \sqrt{y(1-y)} \right)$$

Substituting the values we get

$$x_e = \frac{2Ze^2}{E} = \frac{2Z\hbar c}{E} \cdot \alpha \approx 6.17 \times 10^{-12} \text{ cm.}$$

$$v = \sqrt{\frac{2E}{mc^2}} \cdot c$$

$$mc^2 \approx 4.938 \text{ MeV}$$

(α -particle is 2 protons + 2 neutrons)

$$v \approx 0.0473c$$

$$y \approx 0.162$$

(Note: $\cos^{-1} y \approx \pi/2 - y$)

Thus

$$y \approx 88.8$$

$$R = \frac{v}{2x_0} e^{-y}$$

$$\Rightarrow T \approx 5 \times 10^{17} \text{ sec}$$

(slightly off from Shankar!)

16.2.5 : First note that ($l=0$) spherically symmetric 3d problems can be mapped to 1-d problems with the same potential $V(r)$ where $V(r) = \infty$ for $r < 0$! (as long as $V(r)$ is not singular at the origin). Thus the ^($l=0$) bound states of $V(r) = V_0 + kr$ is the same as the bound states of the corresponding 1-d problem with $V(r) = \infty$ for $r < 0$. Thus the energy levels are given by problem-2!

$$E_n = V_0 + \left\{ \frac{3}{2\sqrt{2}} \left(n + \frac{3}{4} \right) \pi \frac{\hbar k}{\sqrt{\mu}} \right\}^{2/3} \quad \left(\mu \text{ is the reduced mass!} \right)$$

$$\mu = 0.75 \text{ GeV}/c^2$$

Thus

$$E_n = V_0 + \left(n + \frac{3}{4} \right)^{2/3} \frac{\alpha}{\mu^{1/3}} \quad \left(\text{where } \mu \equiv \text{in GeV}/c^2 \right)$$

Thus we have

$$3.1 \text{ GeV} = V_0 + \left(\frac{3}{4}\right)^{2/3} \frac{\alpha}{\sqrt{\mu}} \quad (\psi \text{ mass})$$

$$3.7 \text{ GeV} = V_0 + \left(\frac{7}{4}\right)^{2/3} \frac{\alpha}{\sqrt{\mu}} \quad (\psi' \text{ mass})$$

$$\left(\frac{3}{4}\right)^{2/3} \simeq 0.8255 \quad \left(\frac{7}{4}\right)^{2/3} \simeq 1.4522$$

$$V_0 \simeq 2.3 \text{ GeV}$$

$$\frac{\alpha}{\sqrt{\mu}} = 0.957 \text{ GeV}$$

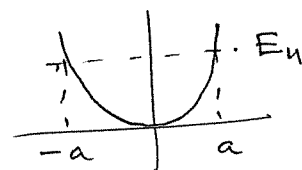
Thus ψ'' mass would be

$$M_{\psi''} = V_0 + \left(\frac{11}{4}\right)^{2/3} \frac{\alpha}{\sqrt{\mu}}$$

$$M_{\psi''} \simeq 4.2 \text{ GeV} \quad \text{in excellent agreement with experiment!}$$

16.2.7 Harmonic Oscillator in the WKB approx: (one-dim).

$$V(x) = \frac{1}{2} m \omega^2 x^2$$



$$E_n = \frac{1}{2} m \omega^2 a^2 \Rightarrow a = \sqrt{\frac{2E_n}{m\omega^2}}$$

$$\int_{-a}^a dx \sqrt{2m(E_n - \frac{1}{2} m \omega^2 x^2)} = (n + \frac{1}{2}) \pi \hbar$$

$$\Rightarrow 2 \sqrt{2m} \sqrt{\frac{1}{2} m \omega^2} \int_0^a dx \sqrt{a^2 - x^2} = (n + \frac{1}{2}) \pi \hbar$$

$$\Rightarrow 2m\omega \int_0^a dx \sqrt{a^2 - x^2} = (n + \frac{1}{2}) \pi \hbar .$$

Set

$$x = a \sin \theta$$

$$dx = a \cos \theta d\theta .$$

$$\Rightarrow 2m\omega a^2 \int_0^{\pi/2} d\theta \cos^2 \theta = (n + \frac{1}{2}) \pi \hbar$$

$$\Rightarrow \underbrace{m\omega a^2}_{\frac{2E_n}{\omega}} \cdot \underbrace{\int_0^{\pi/2} d\theta (1 + \cos 2\theta)}_{\pi/2} = (n + \frac{1}{2}) \pi \hbar$$

$$\Rightarrow \boxed{E_n = (n + \frac{1}{2}) \hbar \omega} \Leftarrow \text{Note these are also the exact energy levels!}$$

WKB approx gives the exact levels for the Harmonic Oscillator!